

- periodično gibanje
perioda t_0

$$x(t+t_0) = x(t)$$

^{2-D} kroženje $\rightarrow$ ^{1-D} sinušno nihanje

$$x = x_0 \sin(\omega t + \sigma)$$

- vsako periodično gib. se da zapisati kot $\sum \sin$
- pogosta v naravi

x ... odmik od (stabilne) ravnovesne lege

x_0 ... amplituda

ω ... krožilna frekvenca [$s^{-1} = \frac{rd}{s}$] $\neq$ Hz!

t_0 ... perioda/nihajni čas

$\nu = \frac{1}{t_0}$... frekvenca [$s^{-1} = \frac{nihaj}{s} = \text{Hz}$]

$\omega t + \sigma$... faza ; fazna perioda = $2\pi \rightarrow \omega t_0 = 2\pi \rightarrow \omega = 2\pi \nu$

σ ... fazni zamik

$$F = ma = m \frac{dv}{dt} = m \frac{d}{dt} \left(\frac{dx}{dt} \right)$$

$$x = x_0 \sin(\omega t + \sigma)$$

$$v = \underbrace{\omega x_0}_{v_0} \cos(\omega t + \sigma)$$

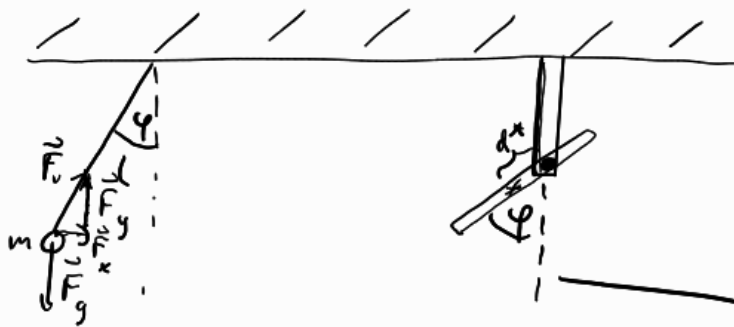
$$\underline{a = -\omega^2 x}$$

$F = -\omega^2 x m$ sila nasp. sorazm. z odklonom

Primer: Hooke $\rightarrow F = -kx \Rightarrow \boxed{k = m\omega^2}$

$$\omega^2 = \frac{k}{m} = \frac{\text{prožnost}}{\text{vztrajnost}}$$

matematično in fizično nihalo



$$M_g = J\alpha$$

$$-mgl \sin \varphi = ml^2 \frac{d^2 \varphi}{dt^2}$$

$$\varphi \ll 1:$$

$$-\frac{g}{l} \varphi = \frac{d^2 \varphi}{dt^2}$$



$$\varphi = \varphi_0 \sin(\omega t + \sigma); \omega^2 = \frac{g}{l}$$

ocena napake:

$$\frac{\sin \varphi}{\varphi} = 1 - \delta; \delta = 0,01$$

$$\frac{\varphi - \frac{\varphi^3}{6}}{\varphi} = 1 - \frac{\varphi^2}{6} = 1 - \delta$$

$$\frac{\varphi^2}{6} = \delta$$

$$\varphi = \sqrt{6\delta} = \sqrt{0,06} \approx \frac{1}{4} \approx 15^\circ$$

$$M_g = J\alpha$$

$$J = J^* + md^{*2}$$

$$-mgd^* \sin \varphi = J\alpha$$

$$\varphi = \varphi_0 \sin(\omega t + \sigma); \omega^2 = \frac{mgd^*}{J}$$

Primer: nihanje noge (hoja brez dolžnjane mase)



$$\begin{aligned} m &= 15 \text{ kg} \\ l &= 1 \text{ m} \end{aligned}$$

$$J = \frac{ml^2}{3} \Rightarrow \omega^2 = \frac{mg \frac{l}{2} \cdot 3}{ml^2} = \frac{3}{2} \frac{g}{l}$$

$$= 15 \text{ s}^{-2} \Rightarrow \omega = 4 \text{ s}^{-1}$$

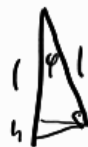
$$t_0 = \frac{2\pi}{\omega} \approx \frac{6}{4} \text{ s} = \underline{1,5 \text{ s}}$$

$$W_k = m \frac{v^2}{2}$$

$$W_{p,g} = mgh = mgl \frac{\varphi^2}{2} =$$

$$= \frac{mgl^2}{l} \frac{\varphi^2}{2} =$$

$$= J \omega^2 \frac{\varphi^2}{2}$$



$$h = l - l \cos \varphi =$$

$$= l(1 - \cos \varphi) =$$

$$\approx l \left(1 - \left(1 - \frac{\varphi^2}{2} \right) \right) =$$

$$= l \frac{\varphi^2}{2}$$

idealno nihalo: $W = \text{konst.}$

dissipativne sile: $\frac{dW}{dt} < 0 \Rightarrow$ nihalo se ustavi
v ravn. legi ($\frac{dW}{dx} = 0$)

$$\vec{F}_{dis} = -\frac{\vec{v}}{v}$$

smerni nasp.
smerni gibanja

Trenje:

$$\vec{F}_{tr} = -K \frac{\vec{v}}{v}$$

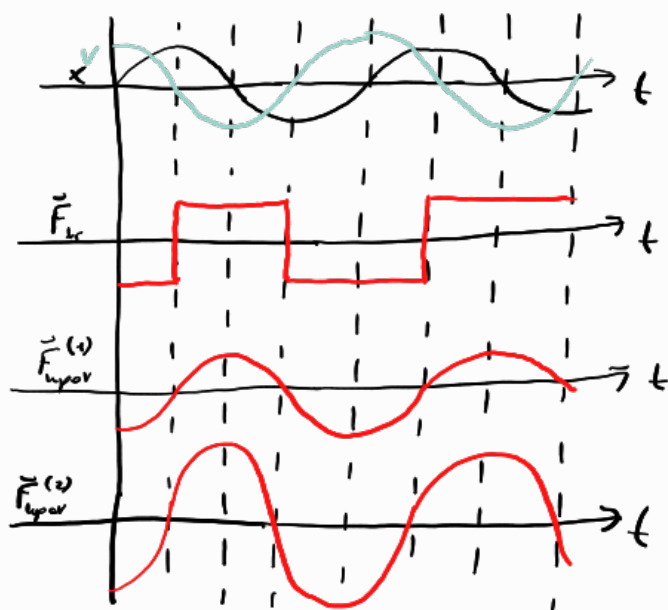
($k_r \cdot \vec{F}_p$)

upor: lin. odv. od hit.

$$\vec{F}_{(1)} = -K_1 v \frac{\vec{v}}{v}$$

kvadr. odv. od hit.

$$\vec{F}_{(2)} = -K_2 v^2 \frac{\vec{v}}{v}$$



$$F_{dis} = -k_1 v$$

$$m a = F_{dis} + F_{vzm} = -k_1 v - kx$$

$$P = \frac{dW}{dt} = F_{dis} \cdot v = -k_1 v^2 \rightarrow \left(\frac{dW}{dt} = -\alpha W \right)$$

relaksacija

$$A = A_0 e^{-\gamma t}$$

$$m a = -2m \gamma v - kx$$

$$\frac{d^2 x}{dt^2} + 2\gamma \frac{dx}{dt} + \omega_0^2 x = 0$$

$$x = A e^{-\gamma' t} \sin(\omega t + \delta)$$

imamo 1 enačbo in 2 prosti parametra (γ' , ω)

$$\dot{x} = A e^{-\gamma' t} [-\gamma' \sin(\omega t + \delta) + \gamma' \omega \cos(\omega t + \delta)]$$

$$\ddot{x} = A e^{-\gamma' t} [\gamma'^2 \sin(\omega t + \delta) - 2\gamma' \omega \cos(\omega t + \delta) - \gamma' \omega^2 \sin(\omega t + \delta) - \omega^2 \sin(\omega t + \delta)]$$

$$\gamma'^2 \sin(\omega t + \delta) - 2\gamma' \omega \cos(\omega t + \delta) - \omega^2 \sin(\omega t + \delta) + 2\gamma' \omega \cos(\omega t + \delta) - 2\gamma' \gamma' \sin(\omega t + \delta) + \omega_0^2 \sin(\omega t + \delta) = 0$$

Ker sta sin in cos neodvisna, lahko razcepimo na 2 dela:

$$\sin(\omega t + \delta) (\gamma'^2 - \omega^2 - 2\gamma' \gamma' + \omega_0^2) = 0$$

$$\cos(\omega t + \delta) (-2\gamma' \omega + 2\gamma' \omega) = 0 \rightarrow \gamma = \gamma'$$

$$-\omega^2 - \gamma^2 + \omega_0^2 = 0 \rightarrow \omega^2 = \omega_0^2 - \gamma^2$$

$$\left. \begin{aligned} F_{dis} &= -2m\gamma v \\ F_u^{(1)} &= -6\pi\gamma r v \end{aligned} \right\} \gamma > \omega$$

$$\omega^2 < 0?$$

$$\omega = i\gamma''$$

$\Downarrow$

$$x = Ae^{-\gamma t} \sin(i\gamma'' t + \delta)$$

$$x = Ae^{-(\gamma + \gamma'')t} + Be^{-(\gamma - \gamma'')t}$$